

Preventive Maintenance

A Practical Implementation Guide for ERP Professionals

Most maintenance teams in D365 are using only 20% of what the Preventive Maintenance module can do. Here's the full picture.

If you've implemented **Dynamics 365 Asset Management** — or you're about to — you've probably discovered that the Preventive Maintenance module is deceptively deep. On the surface, it looks like a scheduler. In reality, it's a **full lifecycle cost and compliance engine** that, when set up correctly, can transform how your maintenance team operates.

This post walks through every concept you need to implement it confidently: maintenance plans, all 11 interval types, counter-based enhancements, schedule runs, work order creation, and cost tracking. With **real-world examples and practical cases** for each concept.

01 Why Preventive Maintenance in D365 Is Worth Getting Right

Unplanned downtime costs manufacturers an average of **\$260,000 per hour** (Aberdeen Group). The promise of a well-configured preventive maintenance program in D365 is simple: *make failures predictable before they become emergencies*.



Maintenance Plans

For individual assets. Linked to an asset, asset type, functional location, or location type. Persists regardless of where the asset is installed.



Maintenance Rounds

For groups of assets needing the same job. One round = one technician route. Perfect for lubrication routes, safety inspections, and fire equipment checks.



The Golden Rule

Use Maintenance Plans when each asset has unique intervals or job types.

Use Maintenance Rounds when 5+ assets need the same job on the same schedule.

Maintenance plan
Name
Monthly
Monthly

Details

PLANNING

Plan date

2/1/2026

Active

Yes

SKIP TOLERANCE

Tolerance days before

0

Tolerance days after

0

DETAILS

Lines

1

Assets

1

Functional locations

Lines

+ Add time line + Add asset counter line Remove

Line	Work order description	Line type	Maintenance job type	Maintenance job type variant	Trade	Finish within days
1.0	Monthly Plan	Time	Tire	Inspection		

02 The 11 Interval Types — Explained Simply

This is where most implementations go wrong. The **Interval Type** field on a maintenance plan line controls everything about *when* a job is scheduled.

Time-Based Interval Types

Finance and Operations
Asset management > Setup > Preventive maintenance > Maintenance plans

Save + New Delete Options

Filter

Monthly

Standard view

Maintenance plans

Maintenance plan

Name

Monthly

Monthly

Details

PLANNING

Plan date

2/1/2026

Active

Yes

SKIP TOLERA

Tolerance day

Tolerance day

Lines

+ Add time line + Add asset counter line Remove

Line	Work order description	Line type	Maintenance job type	Maintenance job type variant	Trade	Finish within hours
1.0	Monthly Plan	Time	Tire	Inspection		

Repeated from plan date

Repeated from start date

Repeated from last work order

Once from plan date

Once from start date

Once reached above (Counter only)

Once reached below (Counter only)

Linked from start date

Linked from last work order

Repeated on aggregated value (Counter only)

Once on aggregated value (Counter only)

Repeated from plan date

Threshold based for example KM, No of Units Produced

Interval Type	Best For	Key Behaviour
Repeated from Plan Date	Fixed-calendar maintenance	Always counts from the Plan Date. Never drift, even if WO finishes late.

Interval Type	Best For	Key Behaviour
Repeated from Start Date	Fleet / multi-asset with different commission dates	Each asset interval starts from its own commissioning date.
Repeated from Last Work Order	Compliance-driven intervals	Restarts from actual WO completion. Guarantees full interval between jobs.
Once from Plan Date	One-time jobs tied to a calendar date	Creates a single schedule line never repeats.
Once from Start Date	Post-installation checks	Creates a single schedule line from asset's start date.
Linked from Start Date	Whichever comes first (time OR counter)	Creates ONE line for the threshold hit first across linked lines.
Linked from Last Work Order	Post-service reset trigger	Same as above but restarts from last WO completion.

✓ Practical Example — Boiler Inspection (Repeated from Last Work Order)

Asset: Boiler B-101 | Interval: 1 Month | Interval Type: Repeated from Last Work Order

The WO was due March 1 but completed March 5 due to a shutdown delay.

Next due date: April 5 (not April 1).

Why this matters: In a regulated industry, the interval must be measured from actual completion — not from a calendar date that may have been missed.

Counter-Based Interval Types

Counter-based lines trigger maintenance based on asset measurements not time. Counters can track *operating hours, kilometers, cycles, temperature, wear thickness*, or any numeric value you register against an asset.

Interval Type	Counter Direction	Real-World Example
Repeated on Aggregated Value	Increasing	Service every 1,000 running hours triggers at 1,000h, 2,000h, 3,000h...
Once on Aggregated Value	Increasing	Major overhaul after 8,000 total hours one time trigger.
Once Reached Above	Increasing (threshold)	Temperature alert: if sensor reads above 18°C, trigger inspection.
Once Reached Below	Decreasing (wear)	Brake pad: replace when thickness drops below 20 mm.

Practical Example — Forklift Brake Pad (Once Reached Below)

Asset: Forklift FL-007 | Counter: Brake Pad Thickness (mm) | Threshold: 20 mm

A technician registers a reading of 17 mm during a routine inspection.

D365 immediately creates a maintenance schedule line for 'Brake Pad Replacement'.

This is reactive maintenance triggered by condition — not calendar.

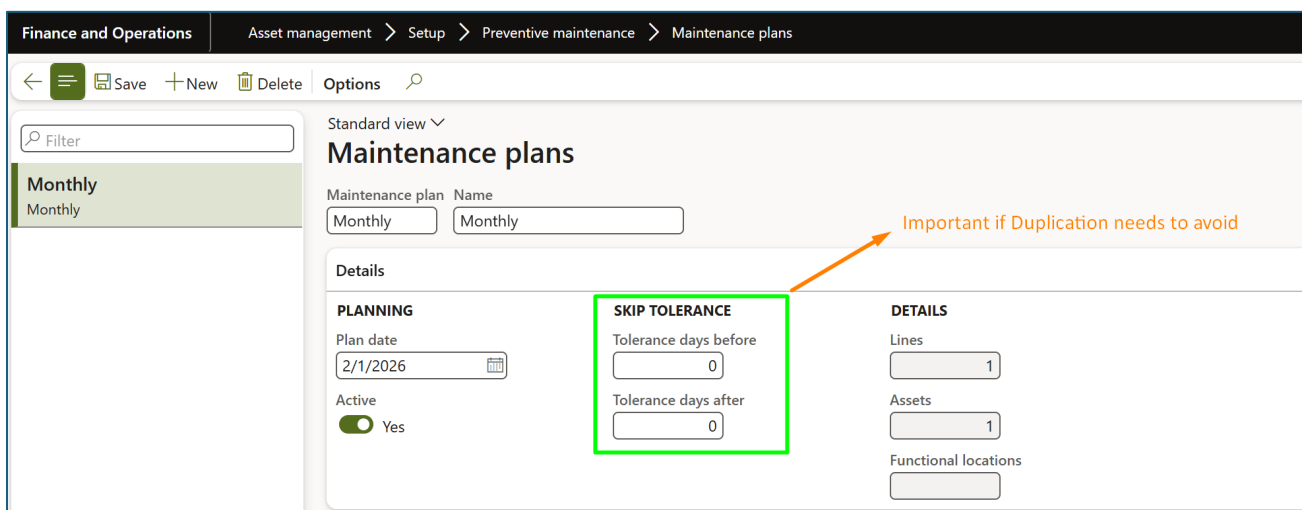
03 Plan Setup — The Fields That Matter

Setting up a maintenance plan takes less than 10 minutes once you know what each field does. Here are the ones that trip people up:

Tolerance Days (Before + After)

This is the most overlooked field in the entire plan setup. If you have a *monthly inspection*, a *6-monthly check*, and an *annual overhaul* all on the same asset they will periodically land on the same date.

Setting Tolerance Days Before = 2 and After = 2 creates a 5-day suppression window around any 'big' job. Smaller overlapping jobs within that window are skipped. Result: clean schedule, no duplication.



The screenshot shows the 'Maintenance plans' setup interface in Dynamics 365 Finance and Operations. The breadcrumb trail is 'Finance and Operations > Asset management > Setup > Preventive maintenance > Maintenance plans'. The left sidebar shows a filter and a list with 'Monthly' selected. The main area is titled 'Maintenance plans' and includes a 'Standard view' dropdown. Below this, there are input fields for 'Maintenance plan' (Monthly) and 'Name' (Monthly). The 'Details' section is divided into three columns: 'PLANNING', 'SKIP TOLERANCE', and 'DETAILS'. The 'SKIP TOLERANCE' column is highlighted with a green box and contains two input fields: 'Tolerance days before' (0) and 'Tolerance days after' (0). An orange arrow points from the text 'Important if Duplication needs to avoid' to the 'SKIP TOLERANCE' section. The 'DETAILS' column contains three input fields: 'Lines' (1), 'Assets' (1), and 'Functional locations' (empty). The 'PLANNING' column contains a 'Plan date' field (2/1/2026) and an 'Active' toggle switch (Yes).

Suppress Overlapping Jobs

Enable this checkbox on the smaller/more-frequent plan lines (monthly, 6-monthly). Leave it OFF on the annual/major job line. D365 will always prioritize the first-listed (largest) job and suppress the rest when they coincide.

Lines

+ Add time line

+ Add asset counter line

Remove

		Period	Period frequency	Suppress overlapping maintenance jobs	Counter	Reset counter
☒		Month	1	☐		☐

Season From / Season To

Underused by most implementers. If you have outdoor equipment, HVAC systems, or seasonal machinery use this to constrain schedule generation to only the relevant months. No more manually deleting off-season work orders.

Step 1

Create Maintenance Plan

Go to: Asset Management > Setup > Preventive maintenance > Maintenance plans

Step 2

Add Time Line

Click "Add time line" on the Lines FastTab

Step 3

Set Interval Type

Select "Repeated from plan date" or "Repeated from start date"

Step 4

Configure Season

Set Season From (start date)
Set Season To (end date)

Configuration Example

Field	Value
Interval type	Repeated from plan date
Period	Year
Period frequency	1
Season from	March 1
Season to	May 31

Result: Maintenance scheduled each year between Mar 1 - May 31

Maintenance plan Name
Vehicle Serv... Vehicle Service

Details

PLANNING	SKIP TOLERANCE	DETAILS
Plan date 7/1/2025	Tolerance days before 0	Lines 1
Active ☒ Yes	Tolerance days after 0	Assets 1
		Functional locations

Lines

+ Add time line + Add asset counter line Remove

	Asset counter	Asset counter time fence in days	Plan date	Service level	Auto create	Work order type	Season from	Season to	Resulting period
☒	☒				☒	Preventive	3/1	4/30	3/1/2026 - 4/30/2026

Pro Setup Tip

Always order your plan lines from LEAST frequent (annual) to MOST frequent (monthly). D365 suppression logic prioritizes the first line in the list. If monthly is listed first and annual is listed second monthly wins. That's the wrong outcome.

04 Schedule & Run What Actually Happens

Running the schedule is the action that turns your maintenance plan configuration into actual work order proposals. Navigate to:

Navigation

Asset Management > Periodic > Preventive Maintenance > Schedule Maintenance Plans

The key dialog fields:

- ▶ **Period + Frequency:** How far ahead to generate lines. 'Month / 3' = 3 months of schedule lines created in one run.
- ▶ **Auto Create if Specified:** If Yes, AND the plan line has Auto Create checked, work orders are created automatically. No manual step needed.
- ▶ **Filters:** Scope the run to a specific plant, location, or asset. Essential during phased rollouts.

Schedule lines with expected start dates **earlier than today** are NOT generated. This is intentionally it prevents the system from flooding your backlog with historical overdue items when you first go live.

Finance and Operations | Asset management > Maintenance schedule > All maintenance schedule

Standard view *

Filter

Expected start

Asset

Maintenance job type

Maintenance forecast hours

Functional location

Reference type

Reference ID

Description

from Service level

plan or round ? May be request raised by users

Plan Name

From job or plan line

Expected start	Asset	Maintenance job type	Maintenance forecast hours	Functional location	Reference type	Reference ID	Description
4/21/2025 5:25:38 PM	FAS-GF-01	FAS-CALIB	100.00	Ground Floor	Maintenance requests	PS-00000002	
4/22/2025 3:00:00 PM	FAS-GF-01	FAS-CALIB	100.00	Ground Floor	Maintenance plans	Repair	Preventive Maintenance Daily based
4/23/2025 3:00:00 PM	FAS-GF-01	FAS-CALIB	100.00	Ground Floor	Maintenance plans	Repair	Preventive Maintenance Daily based
4/24/2025 3:00:00 PM	FAS-GF-01	FAS-CALIB	100.00	Ground Floor	Maintenance plans	Repair	Preventive Maintenance Daily based
4/25/2025 3:00:00 PM	FAS-GF-01	FAS-CALIB	100.00	Ground Floor	Maintenance plans	Repair	Preventive Maintenance Daily based
4/26/2025 3:00:00 PM	FAS-GF-01	FAS-CALIB	100.00	Ground Floor	Maintenance plans	Repair	Preventive Maintenance Daily based
4/27/2025 12:00:00 PM	Elevator_01	Round	10.00	Building A	Maintenance plans	Service	Service of Elevator
4/28/2025 12:00:00 PM	Elevator_01	Round	10.00	Building A	Maintenance plans	Service	Service of Elevator
4/28/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round
4/29/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round
4/30/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round
5/1/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round
5/2/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round
5/3/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round
5/4/2025 3:00:00 PM	FAS-GF-01	Round	10.00	Ground Floor	Maintenance rounds	Daily Round	Daily Round

Automate It

ASSET TYPES

Asset type

Run in the background

Recurrence Alerts

Batch processing

☒ Yes

Task description

Schedule maintenance plans

Batch group

Private

☐ No

Critical Job

☐ No

Monitoring category

Undefined

Start date: 3/3/2026 (01:57:43 pm) (GMT+03:00) Kuwait, Riyadh

OK Cancel

Set the scheduling run as a batch job to execute every Sunday night. Monday morning your planners arrive to a fresh queue of work orders, ready to assign. This single setup decision eliminates the most common complaint from maintenance supervisors: 'I forgot to run the schedule.'

This is where D365 Preventive Maintenance transitions from a *scheduling tool* into a **financial management tool**. Most organisations stop at generating work orders. The cost layer is where the real ROI is.

The Cost Flow



Layer 1 — Job Type Defaults (Pre-WO Forecast)

In Maintenance Job Type Defaults, define the expected labour hours, spare parts, and expenses for each job type. When a work order is created from a maintenance plan, these values auto-populate the work order forecast. Zero manual entry required.

Layer 2 — Work Order Forecast (Per-WO Adjustment)

Each work order carries its own forecast. Before a job starts, the planner can adjust parts quantities, add a one-off contractor cost, or revise labour hours. This is your short-term financial commitment.

Real Example — Pump Overhaul Forecast

Job Type Default auto-populated:

- Labour: 4 hours @ \$75/hr = \$300
- Impeller seal kit (1 unit) = \$150 | Bearing set (2 units) = \$170
- Consumables flat fee = \$40

Total Forecast: \$660

Actual registered after completion:

- Labour: 4.5 hours (corrosion found) = \$337.50
- Parts: \$320 | Expenses: \$45

Total Actual: \$702.50 | Variance: +\$42.50

Layer 3 — Maintenance Budget

Set an annual cost envelope per asset or asset category. Track it against actuals throughout the year. When a work order's cost pushes an asset over budget, the system flags it. This feeds directly into asset replacement decisions — when maintenance cost exceeds a threshold percentage of asset value, it's time to replace.

06 Counter-Based Enhancements — The Hidden Power Feature

Enable **Counter-Based Maintenance Enhancements** in Feature Management and you unlock three capabilities that change how counter maintenance works:

- ▶ **Auto-initialize counters at zero when a new asset is created** — no manual counter entry ever again.
- ▶ **Auto-reset counters when a work order is completed** — enables 'hours since last service' tracking instead of total lifetime hours.
- ▶ **Two new interval types:** Repeated on Aggregated Value and Once on Aggregated Value.

✓ Practical Case — 500-Hour Reset Cycle

Asset: Injection Molding Machine IM-301

Counter: RunningHours (resets to 0 after each WO)

Interval Type: Repeated on Aggregated Value | Period Frequency: 500

Machine runs 500h → schedule line created → WO raised → work done → counter resets to 0.

Machine runs another 500h → next WO. Consistent cycle, forever, regardless of calendar.

07 Full Walkthrough — Packaging Machine Setup

Here is a complete real-world scenario bringing every concept together. A food processing plant needs to configure preventive maintenance for a packaging machine with four different job requirements:

- ▶ Weekly cleaning (time-based, every 7 days)
- ▶ Oil change every 500 operating hours (counter-based)
- ▶ Blade replacement when wear drops below 5 mm (reactive counter)
- ▶ Annual compliance inspection OR every 10,000 units — whichever comes first (linked)

Line	Job Type	Type	Interval Type	Config
Line 1	Cleaning	Time	Repeated from Plan Date	Week / 1 / Auto Create: Yes
Line 2	Oil Change	Counter	Repeated from Start Date	OpHours / 500 / Fence: 365d

Line	Job Type	Type	Interval Type	Config
Line 3	Blade Replace	Counter	Once Reached Below	BladeWear / 5 mm / Reactive
Line 4	Compliance	Time	Linked from Start Date	Year / 1 / Linked with Line 5
Line 5	Compliance	Counter	Linked from Start Date	UnitsProd / 10,000 / First triggers

After linking this plan to asset PKG-001 and running the scheduler (3-month horizon), the system generates: **12 weekly cleaning lines**, projected oil change dates, a standing reactive trigger for blade wear, and a compliance inspection date based on whichever milestone arrives first. All automatically.

✦ Key Takeaways

- ▶ **Interval Type is the most important field in your plan setup.** Match it to your operational reality, not just what's convenient.
- ▶ **Suppress Overlapping Jobs** eliminates duplicate work orders — but only if your lines are ordered correctly (least frequent first).
- ▶ **Counter-Based Enhancements** should be enabled for any asset tracked by meters, hours, or cycles. The auto-reset feature alone saves hours of manual administration.
- ▶ **The cost layer — Job Type Defaults → WO Forecast → Actuals** — is where D365 earns its keep. Use it to build the business case for asset replacement.
- ▶ **Automate the schedule run as a batch job.** A Sunday night batch job means planners start every Monday with fresh work orders.